

Tejas Warake

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PROFESSIONAL SUMMARY

Backend Software Engineer with **2+ years** of experience at **Infineon Technologies** building secure microservices, event-driven pipelines, and agentic AI systems. Cut authentication latency by **35%** via Redis caching, reduced manual workflow effort by **50%** through a multi-agent AI orchestrator, and replaced a **10-minute** manual HSM process with a sub-second automated pipeline. Specializes in distributed systems, applied cryptography, and multi-agent LLM systems.

SKILLS

Languages: Java, Python, Javascript, Typescript, C++, SQL

Backend: Spring Boot, Hibernate, Microservices, Distributed Systems, Kafka, RESTful APIs, WebSockets, OAuth2

Cloud & DevOps: Docker, Kubernetes (K8s), Linux, Jenkins, CI/CD, AWS, Maven, Gradle, Git

Databases: PostgreSQL, Redis, Database Schema Design, NoSQL

AI & Tools: Generative AI, Vector Embeddings, MCP, RAG, LLM Integration, JUnit, Mockito, Postman

EXPERIENCE

Senior Software Engineer | Infineon Technologies

October 2025 – Present | Bengaluru, India

- Designed and deployed **Spring Boot microservices** for secure device tokenization, owning the full backend lifecycle from system design to production deployment.
- Architected an event-driven tokenization pipeline using **Apache Kafka** to replace monolithic APIs, decoupling physical smartcard validation from wallet provisioning to ensure system stability under heavy load.
- Optimized critical authentication APIs with **Redis** distributed caching, reducing p99 latency by **35%** and offloading **PostgreSQL** database reads by **40%**.
- Architected a **multi-agent AI ecosystem** where a central orchestrator delegates complex workflows to specialized sub-agents (Research, Specification, Implementation, and Review), integrating custom tools to autonomously process tasks and reduce manual intervention by almost **50%**.
- Engineered the backend for a conversational AI assistant using **RAG**, processing unstructured enterprise data into vector embeddings to deliver context-aware responses and cut manual data retrieval time by **60%**.
- Developed a **Model Context Protocol (MCP)** server for dynamically fetching API command sets to translate natural language into executable scripts, which reduced manual script generation time by **50%**.

Software Developer | Infineon Technologies

Jun 2024 – September 2025 | Bengaluru, India

- Implemented an automated APDU generation microservice for Mastercard personalization that processes cryptographic keys to build provisioning scripts in **under 1 second**, replacing a **10-minute** manual HSM workflow.
- Engineered secure key vaults for smart card applications, managing **cryptographic** keys across **RSA/ECC** personalization and readout operations for production card issuance.
- Built a **Java security library** with a configurable UI for RSA/ECC key management and digital certificate operations, adopted across multiple customer-facing applications.
- Designed a **multithreaded client tool** to interface with remote test servers, enabling secure, concurrent command execution across physical chip cards and cutting manual test cycle time.

Software Development Intern | Infineon Technologies

July 2023 – May 2024 | Bengaluru, India

- Built a custom VS Code Debug Adapter bridging **GDB** and **DAP**, enabling seamless cross-language step-through debugging for Java and C.
- Developed a **VS Code extension** to read **SFR registers** from a hardware simulator, providing developers with real-time memory state visibility.
- Migrated legacy Ant and Python build scripts to a unified **Maven** configuration, cutting compilation times by **20%** and standardizing the CI/CD pipeline.

PROJECTS

DockFlow – Self-hosted Distributed CI/CD Platform | *Java, Golang, Docker*

[Source Code](#) ↗

- Engineered a CI/CD platform with a **Spring Boot** orchestrator and **Golang** agents running isolated **Docker** pipelines.
- Integrated **WebSockets** for live log streaming and a **PostgreSQL** schema for pipeline metadata and RBAC.

EDUCATION

Vishwakarma Institute of Information Technology

Pune, India

B.Tech in Information Technology; CGPA: 9.39/10

Aug 2020 – May 2024

ACHIEVEMENTS

- **Winner – Infineon Innofest'24:** Developed a functional prototype of the European Digital Identity Wallet.
- **1st Place – IoT Hackathon:** Designed and implemented an IoT-based flood detection and prevention system.
- **LeetCode Knight:** Solved 1000+ Data Structures and Algorithms problems (Top 2% Globally).